BSDS (2025 Semester II): Statistics II

Endsem: Part B

NAME :	**ROLL :**

Time: 90 minutes **Total marks:** 65 **Maximum attainable marks:** 55

Answer all questions.

1. Let X_n, $n \geq 1$ be a sequence of random variables of the following form:

$$P(X_n = k) = \left(1 - \frac{1}{n}\right)\frac{1}{N}, \quad \text{for } k = 1, \ldots, N, \quad \text{and} \quad P(X_n = \sqrt{n}) = \frac{1}{n}.$$

Show that $X_n \xrightarrow{d} Y$ as $n \to \infty$, where Y is distributed as a discrete uniform distribution supported on $\{1, \ldots, N\}$. Without loss of generality, you may assume $n > N^2$. [10]

[*Hint: You may solve using the following steps:*

i. Derive distribution functions of X_n and Y, say, F_n and F_y, respectively.

ii. Identify the set of discontinuity points of F_Y, say $\mathcal{D}$.

iii. For each $y \in \mathbb{R} \setminus \mathcal{D}$, show that $F_n(y) \to F_Y(y)$ as $n \to \infty$.]

2. Let $X_1, \ldots, X_n$ be i.i.d. random variables from `unform`$(0,1)$ distribution. Find the distribution of $Y_n = \sum_{i=1}^n \log(1/X_i)$.

[You may refer to the MGF table provided in the back of the question-answer booklet.] [7]

3. Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim}$ `location-exponential`(α, θ_0), with p.d.f.

$$f_X(x) = \theta_0 \exp\{-\theta_0(x-\alpha)\}, \quad x \geq \alpha,\ \theta_0 > 0 \text{ is known},\ \alpha \in \mathbb{R}.$$

(a) Find the MLE of α.

(b) Given that the MLE of α is a complete and sufficient statistics (CSS), can you find the UMVUE of α?

(c) If the MLE is different from the UMVUE, then identify the better estimator among these two, by comparing their MSEs. [5 + 6 + 4]

4. Let $X_1, \ldots, X_n$, $n = 1, 2, \ldots$, be i.i.d. samples from F_μ with mean μ and standard deviation $\sigma = 3$. We want to set a confidence interval for μ of the form $[\bar{X}_n - 0.3, \bar{X}_n + 0.3]$.

   (a) Using CLT find the minimum sample size n such that the confidence coefficient is at least 0.95. *[Refer to the normal table provided at the end of the question-answer booklet.]*

   (b) If you use the Chebyshev inequality instead, what would be the minimum sample size? [6 + 5]

5. Consider the double exponential distribution with parameter θ and pdf

$$f_X(x;\theta) = (2\theta)^{-1}\exp\{-|x|/\theta\}, \quad x \in \mathbb{R}, \quad \theta > 0.$$

Find the UMP size- α test for testing $H_0 : \theta \leq \theta_0$ against $H_A : \theta = \theta_1$ based on **one sample** X, where $\theta_1 > \theta_0$. [14]

[*Hint: You may go through the following steps:*

i. Find the MP level-α test $\phi^\star$ for testing $H_0^\star : \theta = \theta_0$ against H_A, using Neyman-Pearson's lemma.

ii. Find the power function of the test $\phi^\star$ $\beta_{\phi^\star}(\theta)$, and show that $\beta_{\phi^\star}(\theta)$ is monotone in θ.

iii. Hence, show that the size of the test $\phi^\star$ under H_0, i.e., $\sup_{\theta\leq\theta_0}\beta_{\phi^\star}(\theta)$, is highest at $\theta = \theta_0$ and is α.]

6. Let X be a Pareto distributed random variable with pdf

$$f_\theta(x) = \alpha_0 \frac{\theta^{\alpha_0}}{x^{\alpha_0+1}}, \qquad x > \theta > 0,$$

and $\alpha_0 > 0$ is known.

(a) Find an appropriate pivot $T_X(\theta)$, which is a function of X and θ.

(b) Find (a, b) such that $P_\theta(T_X(\theta) < a) = \alpha/2$ and $P_\theta(T_X(\theta) > b) = \alpha/2$.

(c) Hence, find a 95% confidence interval for θ. [2 + 3 + 3]

Additional Information

Distribution	**Parameters**	**MGF** $M_X(t)$	Range of t
Normal	$\mu \in \mathbb{R},\ \sigma^2 > 0$	$\exp\big(\mu t + \sigma^2 t^2/2\big)$	$t \in \mathbb{R}$
Exponential	$\lambda > 0$	$\lambda/(\lambda - t)$	$t < \lambda$
Gamma	$\alpha,\ \beta > 0$	$(1 - \beta t)^{-\alpha}$	$t < 1/\beta$
Double Exponential	$\mu \in \mathbb{R},\ b > 0$	$\exp\{\mu t\}/(1 - b^2 t^2)$	$\lvert t \rvert < 1/b$

Table 1: MGF of some common distributions.

$\boldsymbol{z}$	$\boldsymbol{\Phi(z)}$
0.900	0.816
0.925	0.822
0.950	0.829
0.975	0.835
1.280	0.900
1.440	0.925
1.645	0.950
1.960	0.975

Table 2: Table of standard normal CDF.